Templates beat LLM-written graph queries two to one; most of Ariadne's other headline numbers were properties of the question set

brendondgr • 2026-08-20 • Status: **Draft**

Slug	`ariadne-project-briefing`
Related	builds-on: `EXP-2026-07-001`–`EXP-2026-08-008` (see Appendix A) • ledgers: `CLAIMS.md`, `DECISIONS.md`, `OPEN_QUESTIONS.md`
Artifacts	repo `Ariadne@4773d4f` • runs `docs/evaluation/` • data MITRE ATT&CK 19.0, MITRE CAPEC 3.9 • figures from `data/briefing.json`

1 Summary

Ariadne answers security-analyst questions in English by querying two graph databases in their own query languages. Nineteen recorded experiments asked one underlying question: are its hand-written query templates worth their cost?

One result held up. Templates beat letting a language model write the database query. On two question sets that share no entity, the template cascade scored 0.9467 and 0.9417. An arm in which the model writes the Cypher or Gremlin itself scored 0.5000 and 0.4183 (Fig. 1, $n = 30$ per cell). Quadrupling that arm's token budget moved it to 0.5367.

Most other headline numbers did not hold up. The near-perfect 0.9972 on the 120-question development catalogue fell to 0.689 on 30 questions written without sight of the code. Two software defects and one flaw in the question set account for the fall. Repaired, those same 30 questions score 0.9350 (Fig. 2).

The claim that templates hold the accuracy failed outright. An arm with every template disabled scored level with or above the cascade on all three question sets. What separates the two is refusal quality: the cascade shows the query behind a refusal on 20 of 21 catalogue refusals, against 11 of 20 (Table 4).

Read the caveats. No result here is measured against a system outside this repository. The merged-graph comparison, which the funding pitch rests on, has never been built.

2 Background

2.1 What the system is

Ariadne answers questions about attacker behaviour by querying two live graph databases and joining the results. MITRE ATT&CK sits in Neo4j and is queried in Cypher. MITRE CAPEC sits in a TinkerPop Gremlin Server and is queried in Gremlin. The two are joined on shared ATT&CK technique identifiers. Every sentence in an answer carries a citation to a retrieved record and to

the query that produced it.

The pipeline plans a question into sub-questions, routes each to a source, retrieves, verifies each claim against the retrieved records, and composes an answer. Almost all of that is deterministic. Query construction fills a fixed library of parameterised, read-only templates rather than generating query text.

2.2 Why the project exists in this form

The design commitment is unfashionable. A reviewer who expects an agent sees hand-written templates and pattern-matched intent classification. The project's own external audit concluded that the way to defend this is to stop selling the mechanism and start selling the measurement (aud, 2026).

That reframing has support in the literature. Writing graph queries from English is measurably hard. Zero-shot execution accuracy on an unseen production graph averages 0.036 across 11 models (Ranganath and Raghavendra, 2026). Zero-shot execution accuracy on ISO-GQL is at or below 4% (Lyu et al., 2026). A peer-reviewed cross-dialect benchmark reports 35.40% average execution accuracy (Su et al., 2026).

Choosing templates over generation is not new either. It is a named family in knowledge-base question answering since 2015 (Bast and Haussmann, 2015; Zheng et al., 2018), catalogued as such in the field's own survey (Gu et al., 2022). The modern analogue constrains decoding rather than eliminating it, and it works (Scholak et al., 2021).

2.3 The gap

Two things were missing when this project started, and the audit verified both across seven search formulations.

First, nobody had run a controlled head-to-head of a query-template library against free-form query generation for graph query languages. The comparison is asserted in a relational setting (fin, 2025) and never measured for Cypher or Gremlin.

Second, nobody had reported federated question answering across security knowledge graphs held in separate engines with different query languages. The closest prior work, AgCyRAG, builds that architecture in the same domain and commits to predefined rather than generated queries (Kurniawan et al., 2025). It reports no quantitative evaluation at all. It scoops the design and cannot scoop the result.

2.4 The questions this record set out to answer

1. Does routing to validated templates beat letting a language model write the query?
2. Is the pipeline's near-perfect score a real capability, or an artifact of a question set written alongside the code that answers it?
3. Does querying two engines separately beat merging both sources into one graph?

2.5 Scope

This report covers work recorded between 2026-07-20 and 2026-08-20. It covers one domain, two catalogues, and one served language model. It compares Ariadne against versions of Ariadne. No external system has been implemented or run, and question 3 is unanswered.

3 What we did

3.1 Design

Every experiment is an *arm*: one configuration of the same pipeline, run over one question set, graded by one grader. Arms differ in three places and are otherwise identical. Table 1 lists them.

Table 1. The nine configurations on record. Each row changes one or more of three things: who writes the plan, how the database query is constructed, and who writes the prose. Arms A–G are the July 2026 study. Arm H and the query-writing arm were built in August 2026.

Arm	Plan	Query construction	Prose
A	Authored	Templates	Deterministic composer
B	Language model	Templates	Language model
C	Authored	Templates	Language model, on raw evidence
D	Authored	Templates	Language model, on verified claims
E	Language model	Templates	Language model, retrieval loop on
F	Authored	Templates	Language model, retrieval loop on
G	Language model	Templates disabled, 2026-07 primitives	Language model
H	Authored	Templates disabled, schema-derived primitives	Deterministic composer
–	Authored	Language model writes the Cypher or Gremlin	Deterministic composer

The query-writing arm is built to be fair rather than easy to beat. The model receives the real database schema, the resolved seed identifiers, the constraints the safety gate enforces, and the output-column convention the harness expects. A rejected query comes back with the rejection reason attached for another attempt. The model is denied the templates and nothing else.

3.2 Data

Two knowledge graphs, pinned and checksummed: MITRE ATT&CK 19.0 in Neo4j, MITRE CAPEC 3.9 in a Gremlin Server. Four question sets, described in Table 2.

Successive sets are kept disjoint by an exclusion registry that records every anchor any set has

Table 2. The four question sets. Expected answers are computed from the raw source files in every case, never written by hand. An anchor whose expected answer could not be computed was discarded rather than repaired.

Set	*n*	How the questions were written
Development catalogue	120	Written alongside the templates that answer them. Each question records the template it expects.
Blind holdout v1	30	Anchors drawn uniformly by a seeded sampler against a frozen tag. Questions written by a model shown one entity and nothing else.
Generated set 1	30	Same protocol, run end to end by one command, seed 20260819.
Generated set 2	30	Same, seed 20260820. Shares no entity with any earlier set.

ever drawn. A fresh random seed alone does not achieve this: over 59 reseeds against holdout v1, 51% of draws shared at least one anchor (`utils/specialists/evalset/tests/test_make_bank.py`).

3.3 Procedure

Every arm runs through one entry point. That entry point saves the answer to each question, the full retrieved evidence, and every executed query:

```
uv run python -m utils.evaluation.experiment_untruncated -arm ARM -root DIR
```

Grading runs separately over those saved files (`utils/evaluation/judge_answers.py`). Question sets are generated by `utils/specialists/evalset/make_bank.py`. The two multi-phase overnight rounds are driven by `scripts/measure_arms.py` and `scripts/night2_sweep.py`, which write a phase-by-phase state file.

3.4 Measures

Correctness The grader's score from 0 to 1 for one answer against the computed expected answer. Reported as an unweighted mean over a set, with a 95% confidence interval over questions. This is the primary measure.

Verdict The grader's categorical call: CORRECT, PARTIAL or INCORRECT.

Answers with a fabrication The count of answers whose judgment lists at least one statement the grader could not find in the retrieved evidence.

Gold in evidence The fraction of expected identifiers present in the retrieved records. This measures retrieval alone and ignores the answer.

Refusal with proof Of the answers that declined to assert, the share that attach the executed query as justification for declining.

Wall clock Seconds to complete a whole question set, one machine, one run, no parallelism across sub-questions.

3.5 Analysis

Two graders appear in the record. The July study and holdout v1 use one independent Sonnet 5 agent per answer, given the expected assertions and the full retrieved evidence. The August rounds use a locally served model (`relay/skynet@qwen38-27B-awq`) over the same protocol, calibrated against the Sonnet judgments on 30 answers at Pearson $r = 0.9275$, running 0.031 generous on the mean.

Grading the same fixed set of deterministic answers three times gives a spread of 0.0050. That is the noise floor every difference in the August rounds is read against.

Where a comparison holds the question set constant, it is paired question by question. The answer-writing repair carries a paired percentile bootstrap, 2000 resamples, seed 1729. Differences between configurations are reported treated minus reference, so both effects in the July study are negative and directly comparable.

3.6 Why this design

Judge-scored correctness is primary because the alternatives are self-consistency checks. Three of the four originally specified end-to-end metrics measure whether the system's output agrees with its own retrieved records. A confidently wrong pipeline passes all three. All four read 1.0 while an independent panel scored the same answers at 0.8946.

Grading is cross-model by construction. The first end-to-end evaluation had one model grading its own output and reported faithfulness 0.973 against an independent 0.956, which is the self-preference the concern predicted.

Expected answers are computed from raw source files rather than annotated. An audit of 16 knowledge-graph question-answering datasets found average factual correctness of 57% (Zhang et al., 2025). Computing gold is a direct response to that finding, and it is why a question whose gold cannot be computed is dropped.

4 What we found

4.1 Templates against a model writing the query

On 30 questions generated the same morning and seen by no arm, the template cascade scored 0.9467 (95% CI 0.915–0.972). An arm in which the model writes the Cypher or Gremlin itself scored 0.5000 (95% CI 0.415–0.585). The plan was held constant between them. Query construction was the only variable.

The result replicated on a second set of 30 questions sharing no entity with the first: 0.9417 against 0.4183 (Fig. 1).

Three of that arm's failures on set 1 were the safety gate refusing every attempt, and two of those three were empty replies from the model. Re-running the arm on set 2 with four times the token budget moved it from 0.4183 to 0.5367 (95% CI 0.447–0.628, $n = 30$). Gate refusals went from 3 to 5. Table 3 gives the full breakdown of where its questions ended up.

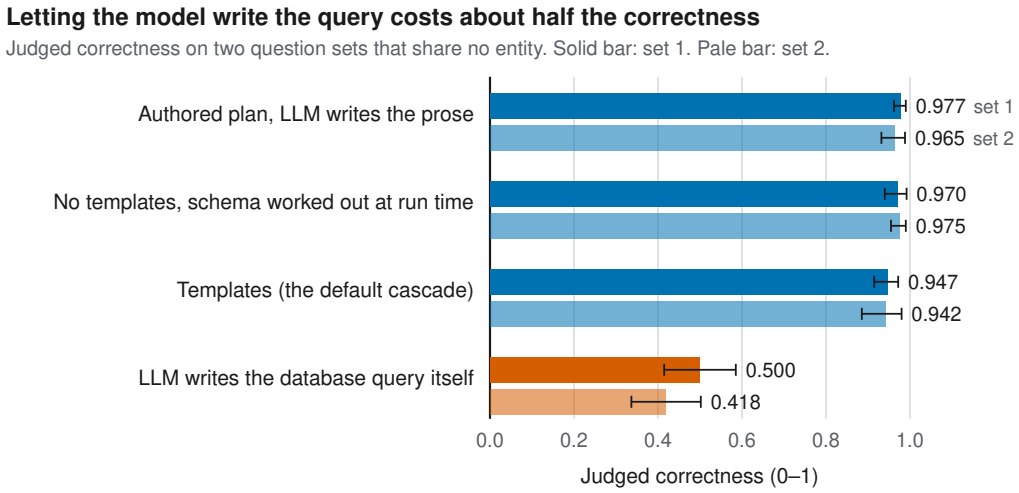


Figure 1. Letting the model write the query costs about half the correctness, on two question sets that share no entity. Judged correctness for four configurations, $n = 30$ per cell except the prose-writing arm on set 2, where one case failed to grade and $n = 29$. Whiskers are 95% confidence intervals over questions. Both sets were graded by the same locally served grader. Produced by `figures/make_figures.py`.

Table 3. Where the query-writing arm's questions ended up, $n = 30$ per column. The second row counts questions whose query ran and returned rows that our own evidence reader could not interpret. That reader dispatches on the column layouts the templates emit, so the failure is ours rather than the model's. Produced by `tables/make_tables.py`.

Outcome	Set 1	Set 2	Set 2, 4× budget
Query ran and produced usable evidence	19	16	20
Query ran; our evidence reader could not parse it	7	8	5
Safety gate refused every attempt	3	5	5
Database refused the query	1	1	0
No result recorded	0	0	0

Crediting every one of those reader failures as a success raises the arm from 0.5000 to 0.5739 on set 1 ($n = 23$ scorable).

4.2 The same system, six defensible numbers

The deterministic cascade has been graded on three question sets and at three code states. One grader scored all six measurements. The values range from 0.7200 to 0.9833 (Fig. 2).

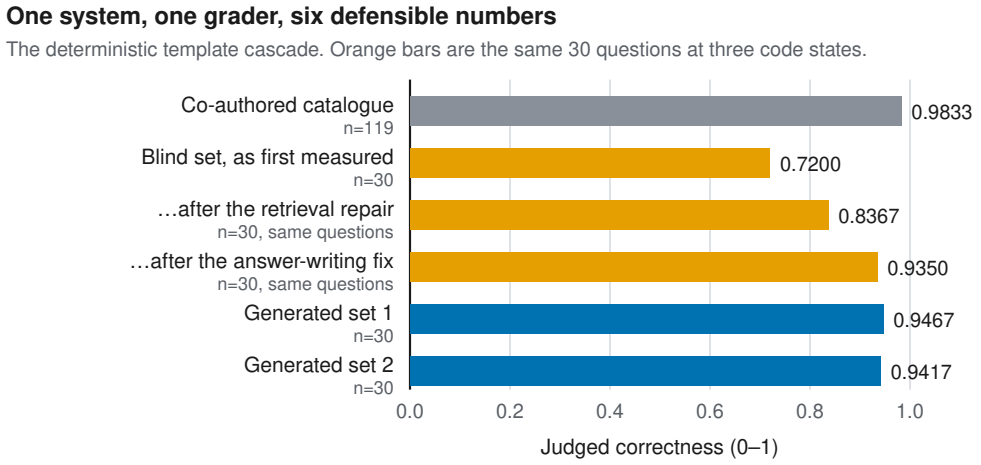


Figure 2. The deterministic template cascade scores anywhere from 0.72 to 0.98 depending on which question set and which code state is quoted. Orange bars are the same 30 blind questions at three code states, so those three are paired. The grey and blue bars are different question sets and are not paired with anything. All six were scored by the same locally served grader; an independent Sonnet 5 panel scored the second bar's answers at 0.689 rather than 0.7200. Produced by `figures/make_figures.py`.

The first blind measurement is the sharpest single contrast in the record. Arm A scored 0.9972 on the co-authored catalogue ($n = 120$) and 0.689 on the blind set ($n = 30$). Arm F scored 0.9957 and 0.7083. The fall was concentrated rather than spread: 1.00 on seven of sixteen question categories, 0.00 on the category asking which groups use a technique.

4.3 Where the blind loss sat, and what removed it

Retrieval accounted for the first part. Gold-in-evidence for the cascade on the blind set was 0.7688. Two repairs raised it to 0.8952 and then to 1.0000, with the 120-question catalogue unmoved at 1.0000 and 0 of 120 questions regressed.

The remainder sat in answer writing. Two defects were fixed together. The prose named at most three items when more had been retrieved and verified. The prose was also never itself checked against the verified claim set. Re-graded on the same 30 questions, correctness went from 0.8367 to 0.9350, a paired difference of +0.0983 (95% CI +0.025 to +0.180, 10 questions improved, 2 regressed, 18 unchanged).

The gap between the co-authored catalogue and the blind set was 0.1472 before those fixes and 0.0483 after them, on the same questions.

4.4 Turning the templates off

An arm with every hand-written template disabled, using only schema-derived primitives, scored at or above the cascade on all three blind question sets. The figures are 0.8483 against 0.8367 on the blind holdout, 0.9700 against 0.9467 on generated set 1, and 0.9750 against 0.9417 on generated set 2 ($n = 30$ each). On the catalogue it scored 0.9692 against 0.9839 ($n = 120$).

The two configurations separate on fabrications. On generated set 1 the no-templates arm produced 2 answers containing a statement not in the evidence, against 0 for the cascade. On the catalogue the counts were 9 and 3 out of 120.

An earlier version of the no-templates arm, before the schema-derived capabilities existed, scored 0.6388 on the catalogue.

4.5 Refusals that carry their query

Table 4 gives every measurement of refusal quality on record.

Table 4. How often a refusal to answer arrives with the executed query attached. Counts are of answers that declined to assert. The first four rows predate the answer-writing fix; the last eight follow it. Produced by `tables/make_tables.py`.

Question set	Configuration	Refusals with proof	Rate
Co-authored catalogue	Templates	20 of 21	95%
Co-authored catalogue	No templates	11 of 20	55%
Blind holdout v1	Templates	3 of 3	100%
Blind holdout v1	No templates	2 of 4	50%
Generated set 1	Templates	4 of 4	100%
Generated set 1	No templates	4 of 4	100%
Generated set 1	LLM writes the prose	4 of 4	100%
Generated set 1	LLM writes the query	18 of 20	90%
Generated set 2	Templates	5 of 5	100%
Generated set 2	No templates	4 of 4	100%
Generated set 2	LLM writes the prose	5 of 5	100%
Generated set 2	LLM writes the query	16 of 19	84%

On the catalogue the cascade attached the query to 20 of 21 refusals and the no-templates arm to 11 of 20. On the blind set the counts were 3 of 3 and 2 of 4. After the answer-writing fix both arms reach 4 of 4 and 5 of 5 on the two generated sets.

4.6 Repeat runs

Two configurations were run three times over the same 30 questions (Table 5). The deterministic cascade produced byte-identical answers on all 30 questions in both repeats. The arm with a language model writing the prose produced 6 identical answers out of 30 in one repeat and 3 out of 30 in the other.

Table 5. Answer text compared character by character against the first run, over the same 30 questions. Produced by `tables/make_tables.py`.

Configuration	Repeat run	Answers identical to run 1
Templates	`repeat-deterministic-r2`	30 of 30
Templates	`repeat-deterministic-r3`	30 of 30
LLM writes the prose	`repeat-gold_loop_llm-r2`	6 of 30
LLM writes the prose	`repeat-gold_loop_llm-r3`	3 of 30

Graded, the three runs of the prose-writing arm scored 0.9490, 0.9750 and 0.9773, a spread of 0.0283 over identical inputs. Answers containing a fabrication across those three runs numbered 0, 2 and 1.

4.7 A second grader over the same saved answers

A different model re-graded the saved answer files from generated set 1. The files were identical, so every difference belongs to the grader (Table 6).

Table 6. Two graders over one set of saved answers. *Offset* is our grader's mean minus the second grader's mean. r is the Pearson correlation between the two graders' per-question scores. The second grader emitted only two distinct scores, 0.5 and 1.0, which floors r on the three high-scoring rows. Produced by `tables/make_tables.py`.

Configuration	n	Our grader	Second grader	Offset	r
Templates	30	0.9467	0.9833	−0.0367	−0.01
No templates	29	0.9690	0.9828	−0.0138	+0.17
LLM writes the prose	30	0.9773	0.9833	−0.0060	+0.37
LLM writes the query	27	0.5241	0.7407	−0.2167	+0.67

The offset ran from −0.0060 to −0.2167 across the four configurations, and it shrank in the order our own grader ranks them.

4.8 The seven-configuration study of July 2026

Table 7 gives the original study, all on the co-authored catalogue.

Three contrasts came out of it. Substituting a language-model planner for the authored plan cost −0.1185 (arm B minus arm C). Disabling the templates cost −0.3568 (arm G minus arm F), a ratio of 3.0 between the two effects. Binding the prose writer to the verified claim set rather than to raw evidence moved correctness from 0.9690 with 8 PARTIAL verdicts to 0.9917 with 0.

The deterministic arm completed all 120 questions in 1.6 s against 1019.7 s for the language-model-planned arm, a ratio of 637.

4.9 What did not run

The merged-graph comparison did not run. Loading both sources into one Neo4j instance stopped part-way: ATT&CK loaded, CAPEC's nodes loaded, and CAPEC's relationships did not.

Table 7. The July 2026 study, 120 co-authored questions per arm, graded by an independent Sonnet 5 panel. Arm G is the pre-2026-08 no-templates control. Wall clock is one machine, one run, no parallelism across sub-questions. Produced by `tables/make_tables.py`.

Arm	Configuration	Correctness	Answers with a fabrication	Wall clock (s)
A	Deterministic	0.9972	0	1.6
B	LLM planner	0.8505	8	1,019.7
C	Writer on evidence	0.9690	3	583.8
D	Writer on claims	0.9917	1	501.0
E	LLM planner + loop	0.8560	9	1,043.1
F	Authored plan + loop + writer	0.9957	3	533.4
G	Templates disabled	0.6388	7	914.8

The loader admits relationship names matching `^[A-Z][A-Z0-9_]*$`, and six of CAPEC's sixteen relationship names come from the source XML in mixed case. The store holds 9138 nodes and ATT&CK's 27669 relationships, and must not be queried.

One case in generated set 2 has no grade. Three grading attempts at rising token budgets returned two empty replies and one gateway error. That arm reads $n = 29$.

5 What it means

5.1 The finding, in one sentence

Deciding which query to run is where this design earns its keep. Holding the templates and writing the prose are not.

5.2 Interpretation

The query-writing gap is real and it survives its fairest corrections. Two disjoint question sets put the same gap in the same place. Quadrupling the token budget did not close it, and gate refusals went up rather than down under the larger budget. Crediting every failure that belongs to our own evidence reader still leaves the arm at roughly half the cascade. The mechanism behind the gap was not isolated. The failure split separates gate refusals from reader failures, and no experiment attributes the residue to a specific weakness in the generated queries. [confidence: strong]

The claim that templates hold the accuracy is dead as stated. It was described in the claims ledger as the strongest result in the study, and it rested on a no-templates control that predates the schema-derived capabilities. Rebuilt, that control matches or beats the cascade on every question set it has been graded on. The ratio of 3.0 in Table 7 prices a missing capability as much as it prices hand-authoring. [confidence: strong]

What the templates do still hold is less than it first looked. The refusal-quality gap in Table 4 was wide on the catalogue and on the blind set, and it disappeared on both generated sets once the answer-writing fix landed. The most likely reading is that the earlier gap was a defect in the generic path rather than a property of templates. That reading was not tested directly. The one property the templates hold without qualification is reproducibility: identical answers on repeat runs, so any difference the cascade shows against another configuration is a real difference. [confidence: suggestive]

The question set moved the numbers as much as the code did. The same system spans 0.72 to 0.98 across the six measurements in Fig. 2. Two-thirds of what was first attributed to the question set turned out to be software defects that hit naturally-phrased questions harder than catalogue questions. A bias measurement taken over a defective pipeline measures the defect too. [confidence: established]

The top three configurations are not separable. The prose-writing arm varies by 0.0283 against itself over identical inputs. Every difference among the top three arms on either generated set is smaller than that. The order of the top two also swapped between the two sets. Any table that ranks those three describes one run of one question set. [confidence: strong]

5.3 How this fits what was already known

The size of the query-writing gap is consistent with the published difficulty of the task. Zero-shot execution accuracy on unseen production graphs averages 0.036 across 11 models (Ranganath and Raghavendra, 2026), and a peer-reviewed cross-dialect benchmark reports 35.40% (Su et al., 2026). Our arm scores well above both, which is what a good-faith build given the schema, the seeds and a repair loop should do. It is still half the cascade.

Nothing here makes the mechanism novel. Choosing templates over generation is a named family in knowledge-base question answering (Bast and Haussmann, 2015; Zheng et al., 2018; Gu et al., 2022), and the modern constrained-decoding line is stronger and better evaluated (Scholak et al., 2021). AgCyRAG independently built this architecture in this domain and wrote down the same design argument, with no quantitative evaluation of any kind (Kurniawan et al., 2025). The contribution available to this project is the measurement, not the design.

The grading protocol is a live vulnerability. Judge-based evaluation in this area has been shown to swing win rates by more than 30% from position alone (Zeng et al., 2025). Our protocol is better than the one that critique dismantles: a different model family, given the expected assertions and the full evidence, one agent per answer. There is still no agreement statistic against a human.

5.4 Limitations, ranked by what they would change

1. **No external baseline exists.** Every comparison in this report is Ariadne against Ariadne. No result here may be described as beating anything outside this repository. This blocks the federation claim entirely, since the merged-graph counterfactual has never been built.

2. **Both graders are language models, neither validated against a human.** The second grader in Table 6 emitted two distinct scores and cannot resolve differences of a few hundredths. It also ran less generous on our grader's weakest arm than on its strongest, which is the pattern that would matter if it held. This changes every absolute correctness value in the report. It does not close the two-to-one gap, which survives the largest offset the second grader produced.
3. **Thirty questions per blind set.** A difference of a few hundredths is noise at that size, which is exactly the regime the top three configurations sit in.
4. **One domain, one served model, one span of days.** Nothing here generalises to another model without re-running, and the served model's quantisation is recorded as unknown rather than guessed.

5.5 Confidence roll-up

Templates beat a model writing the query, at roughly two to one, on anchored questions in this domain. [confidence: strong] The near-perfect catalogue figure was substantially an artifact of a co-authored question set. [confidence: established] The templates are not where the correctness lives. [confidence: strong] Federation beats merging. [confidence: unknown]

6 What's next

6.1 One decision is blocking

The merged-graph comparison cannot proceed until someone chooses between two repairs to the Neo4j loader, and the choice is a design judgment rather than a bug fix.

Option A widens the relationship-name rule to match the label rule already beside it. One character class. The merged graph then names relationships exactly as the CAPEC source does.

Option B renames CAPEC's six mixed-case relationships to the loader's existing convention. The safety rule stays untouched. The merged store then names its relationships differently from the Gremlin store holding the same data, so a query written against one does not port to the other.

Option B quietly handicaps the merged arm in the comparison it exists to inform. Option A touches a function named for safety. Whichever is chosen, the merged store must be wiped and reloaded from empty, since it currently holds half a catalogue.

6.2 Recommendations this report supports now

Stop quoting a single arm ranking. The top three configurations are inside the prose-writing arm's own run-to-run variation, and their order swapped between the two generated sets. Report them as not separable, or run each arm three times and report the spread.

Re-grade with a capable independent model. Do this before anything leaves the repository. The second grader in Table 6 emitted two distinct scores and cannot resolve this range. The earlier Sonnet 5 panel can. Roughly one grading round.

Record the second night in the research record. The repeat runs, the second question set,

the token-budget re-run and the cross-grading exist only as a sweep output directory. By this project's own rules they do not count until they are scaffolded as an experiment.

Make question identifiers carry their question set. The two generated sets reuse one identifier scheme over different entities. Nothing is wrong today, since every roll-up scopes by directory. It has already produced one false diagnosis in a report.

6.3 Open questions this report did not settle

Where is the coverage cliff? Every question in every set names an entity. Nothing measures the regime where a template router is weakest, so the design's price on open-ended questions is unreported. Roughly 40 non-anchored questions would price it.

Does the verifier do anything? No experiment disables the verification gates one at a time. Fabrication counts sit near zero across arms, and no measurement attributes that to the verifier rather than to the templates.

How does either grader compare with a human? A reviewer will ask for an agreement statistic on a sample of answers. There is none, and the published critique of judge protocols makes this the first question a reader will reach for.

Should the evidence reader be decoupled from the templates? This is the one ambiguity inside the headline result. Either the coupling is an artifact of our harness, and the query-writing arm should be measured without it. Or guaranteeing consumable output is a genuine property of templates, and that arm should wear the cost. This report gives both numbers. A paper has to pick one reading and defend it.

References

FinAI data assistant: LLM-based financial database query, 2025. arXiv:2510.14162. Preprint; results body not retrieved — cited for the existence of the comparison, not for figures.

External research audit of the ariadne project. Technical Report audit-2026-08-03, Ariadne research record, 2026. docs/research/audit-2026-08-03.md.

Hannah Bast and Elmar Haussmann. More accurate question answering on freebase. In *CIKM*, 2015. doi: 10.1145/2806416.2806472.

Yu Gu, Vardaan Pahuja, Gong Cheng, and Yu Su. Knowledge base question answering: A semantic parsing perspective. In *AKBC*, 2022. arXiv:2209.04994.

Kabul Kurniawan, Rayhan Firdaus Ardian, Elmar Kiesling, and Andreas Ekelhart. AgCyRAG: an agentic knowledge graph based RAG framework for automated security analysis. In *RAGE-KG 2025 @ ISWC 2025, CEUR-WS Vol-4079, paper 11*, 2025. https://ceur-ws.org/Vol-4079/paper11.pdf, accessed 2026-08-03.

Songlin Lyu, Lujie Ban, Zihang Wu, Tianqi Luo, Jirong Liu, Chenhao Ma, Yuyu Luo, Nan Tang, Shipeng Qi, Heng Lin, Yongchao Liu, and Chuntao Hong. Text2GQL-Bench: A text to graph query language benchmark, 2026. arXiv:2602.11745. Preprint, accessed 2026-08-03.

Suraj Ranganath and Anish Raghavendra. PIPE-Cypher: Automatic enterprise benchmark

generation for text-to-cypher systems, 2026. arXiv:2606.08481. Preprint, accessed 2026-08-03.

Torsten Scholak, Nathan Schucher, and Dzmitry Bahdanau. PICARD: Parsing incrementally for constrained auto-regressive decoding from language models. In *EMNLP*, pages 9895–9901, 2021.

Yanning Su, Yuhang Zhou, Yang Fang, Sen Liu, Guangnan Ye, and Hongfeng Chai. GQLBench: A large-scale cross-domain, cross-dialect benchmark for NL2GQL. In *ACL (Volume 1: Long Papers)*, pages 31989–32014, 2026.

Qiming Zeng, Xiao Yan, Hao Luo, Yuhao Lin, Yuxiang Wang, Fangcheng Fu, Bo Du, Quanqing Xu, and Jiawei Jiang. How significant are the real performance gains? an unbiased evaluation framework for GraphRAG, 2025. arXiv:2506.06331. Preprint, accessed 2026-08-03.

Liangliang Zhang, Zhuorui Jiang, Hongliang Chi, Haoyang Chen, Mohammed Elkoumy, Fali Wang, Qiong Wu, Zhengyi Zhou, Shirui Pan, Suhang Wang, and Yao Ma. Diagnosing and addressing pitfalls in KG-RAG datasets: Toward more reliable benchmarking. In *NeurIPS Datasets and Benchmarks Track*, 2025. arXiv:2505.23495.

Weiguo Zheng, Jeffrey Xu Yu, Lei Zou, and Hong Cheng. Question answering over knowledge graphs: Question understanding via template decomposition. *PVLDB*, 11(11):1373–1386, 2018.

A Every experiment on record

Table 8 is parsed from the research record's own generated index, which is rebuilt from the experiment manifests by `make index`. Nothing in it is retyped.

B The claims ledger as it stands today

Table 9 is parsed from `docs/research/CLAIMS.md`. Claim text is truncated; the ledger carries the full wording and the evidence trail.

Two rows in that table disagree with their own notes and should be reconciled before anything is written from it.

C-003 reads `supported` while its notes record that the no-templates arm judged level with the cascade and that the claim needs rewriting. On the evidence in §4.4 the status is at best `partial`.

C-012 reads `supported` on correctness. Its wording also claims a cost advantage, and no cost figure was taken in the run that supports it.

C Reproducing the numbers in this report

One script extracts every value in the figures and tables from the experiments into one file. The figure and table generators read only that file:

Table 8. The nineteen experiments this report draws on. A superseded entry is kept deliberately, so that a dead end is walked into once rather than twice. Produced by `tables/make_tables.py`.

ID	Title	Status	Primary metric
EXP-2026-07-001	Arm A — fully deterministic pipeline (planning, extraction, synthesis)	complete	correctness 0.9972 (n=1)
EXP-2026-07-002	Arm B — LLM planning with deterministic extraction and an LLM writer	complete	correctness 0.8505 (n=1)
EXP-2026-07-003	Arm C — authored plan with an LLM writer reading raw evidence	complete	correctness 0.9690 (n=1)
EXP-2026-07-004	Arm D — authored plan with an LLM writer reading verified claims	complete	correctness 0.9917 (n=1)
EXP-2026-07-005	Arm E — LLM planning with the facet-chasing retrieval loop enabled	complete	correctness 0.8560 (n=1)
EXP-2026-07-006	Arm F — authored plan, facet-chasing loop, LLM writer over verified claims	complete	correctness 0.9957 (n=1)
EXP-2026-07-007	Arm G — generic primitives only, every hand-written template disabled	complete	correctness 0.6388 (n=1)
EXP-2026-07-008	Component-level evaluation of the planner and both query specialists	complete	planner_source_recall 1.0000 (n=1)
EXP-2026-07-009	End-to-end evaluation against the four goals.md pipeline metrics	complete	cross_source_answer_accuracy 1.0000 (n=1)
EXP-2026-07-011	Portability audit — what survives a dataset that is not MITRE	complete	—
EXP-2026-08-001	Is Arm G's gap a capability gap in the schema-driven layer, or domain knowledge?	complete	gold_in_evidence 1.0000 (n=1)
EXP-2026-08-002	Does the deterministic pipeline's near-ceiling correctness survive a blind, uniformly-sampled holdout?	complete	correctness (not recorded)
EXP-2026-08-003	Do the schema-driven primitives generalise to questions nobody wrote templates for?	complete	gold_in_evidence 0.8952 (n=1)
EXP-2026-08-004	Does making the cross-source bridge an outer join close the residual blind-set gap without regressing?	complete	blind_gold_in_evidence 1.0000 (n=1)
EXP-2026-08-005	Is the blind holdout's directional loss a property of these templates, or of a missing schema-driven check?	complete	blind_gold_in_evidence 0.8952 (n=1)
EXP-2026-08-006	Does the outer-join bridge close the residual blind gap in the default cascade without regressing it?	complete	blind_gold_in_evidence 1.0000 (n=1)
EXP-2026-08-007	Judged, at last — does the repaired cascade answer better, and do the generic and LLM-loop arms differ from it on anything that matters?	complete	blind_correctness_cascade 0.8367 (n=30)
EXP-2026-08-008	Do the answer-writing fixes help, how do four arms compare on unseen questions, and does an LLM writing its own queries beat templates?	complete	llm_query_vs_templates_fresh_bank (not recorded)
EXP-2026-07-010	First independent cross-model grading pass (the pre-fix committed baseline)	superseded → EXP-2026-07-001	correctness 0.8946 (n=1)

Table 9. Claim status at 2026-08-20, straight from the ledger. Produced by `tables/make_tables.py`.

ID	Claim	Status
C-001	On an anchored, in-distribution question set, a fully deterministic pipeline — template routing, deterministic extraction, deterministic …	partial
C-002	Substituting LLM query planning for an authored plan is the dominant single source of correctness loss in the pipeline, with extraction and synthesis …	partial
C-003	The hand-written query templates, not the planner, are where the system's accuracy lives: disabling them costs roughly three times what substituting …	supported
C-004	The deterministic path is not merely competitive with the LLM-planned path but orders of magnitude cheaper: the full 120-question catalogue completes …	supported
C-005	Binding an LLM narrator to the *verified claim set* rather than to raw retrieved evidence removes the accuracy-versus-readability trade-off.	supported
C-006	The value of an iterative retrieval loop is a function of plan quality: it compensates for weak planning at substantial cost and does not substitute …	supported
C-007	Constructing queries by filling validated read-only templates places query validity and safety adherence at a ceiling by construction, rather than …	supported
C-008	A deterministic safety gate over templated queries blocks adversarial probes without leakage.	partial
C-009	Self-judged evaluation materially overstates answer quality relative to independent cross-model judging, which is why the primary evaluation must be …	supported
C-010	The system's dataset-specific surface is bounded and countable rather than pervasive: the schema-agnostic machinery transfers, and re-authoring cost …	supported
C-012	Routing to a fixed library of validated parameterised templates outperforms letting an LLM plan and generate the queries, in correctness and in cost.	supported
C-017	Claim-level verification with enforced abstention carrying executed-query proof eliminates unsupported claims without reducing answer coverage.	partial
C-021	Evaluating a graph-question-answering system on questions co-authored with the code that answers them overstates its correctness by a measurable and …	supported
C-011	Federated question answering across cybersecurity knowledge graphs held in *separate engines with different query languages* is a contribution …	unsupported
C-013	The near-ceiling correctness of the deterministic path is a genuine capability rather than an artifact of a question set co-authored with the …	refuted
C-014	The coverage cliff of template routing is measurable: the design's *price* on open-ended, non-anchored questions is bounded and reportable, not …	partial
C-015	Federating across two engines outperforms merging both sources into a single graph and querying that.	unsupported
C-016	Mechanically localising the last pipeline stage at which the gold answer was still recoverable contradicts the assumption that LLM pipeline failures …	partial

```
python3 data/collect.py
python3 figures/make_figures.py
python3 tables/make_tables.py
```

`data/collect.py` raises rather than defaulting when a key is absent, so a figure cannot be built from a metric that has gone missing. It reads each experiment's `data/metrics.json` and the second night's `docs/evaluation/night2-2026-08-20/metrics.json`.

Raw run artifacts are not in the research record. They live under `docs/evaluation/`, and each experiment carries a directory hash in its `data/POINTERS.md` that folds every file's path and content digest in sorted order.

D Known problems with this report

The second night is not yet an experiment. Section 4.6, 4.7, the second generated question set and the token-budget re-run come from a sweep output directory rather than from a scaffolded experiment. By this project's own rules those numbers do not yet count. This report quotes them anyway, labelled, on the grounds that a briefing that omitted them would be less honest than one that flags them.

Two graders are mixed across the record. Table 7 and the first blind measurement come from a Sonnet 5 panel. Everything from 2026-08-19 onward comes from a locally served grader calibrated against it. Figure 2 uses the local grader throughout for that reason, which is why its first blind bar reads 0.7200 rather than the Sonnet panel's 0.689.

Status. Draft. No second person has worked through the checklist. Reviewer and date: none.

E Correction, 2026-08-21

§4.9 of this report says the merged-graph comparison did not run, §5 lists its absence as the first limitation, and the confidence roll-up tags federation [confidence: unknown]. **Those statements were true when written and are now out of date.** They are corrected here rather than edited above, because a report that silently rewrites itself is not a record.

The counterfactual was built and measured the next day (`EXP-2026-08-009`). One Neo4j holding both catalogues, answered by one Cypher specialist, compared against the same primitives and the same deterministic composer over two stores — no language model on either side, so both are byte-reproducible. Merging scored −0.0334 against federating on one blind question set and ±0.0000 on the other, $n = 30$ each, with intervals overlapping almost entirely.

So **federation does not beat merging on correctness**, and C-015 is `refuted` as written. What survives of the federation argument — provenance, licensing, staleness, not re-running a merge when a catalogue updates — is untested and has never been written down as a claim.

One further correction to §5: that section says every comparison in the report is Ariadne against Ariadne. It still is. The merged arm is another configuration of the same system, so it does not close the no-external-baseline gap.